Minimum Drawdown and Catastrophic Elevations for Lake Powell

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This report was generated on: June 8 , 2026

Overview

This is an R Markdown document. The document generates a plot showing an example Minimum Drawdown and Catastrophic Elevations for Lake Powell. The plot also includes:

1. A trace showing Lake Powell storage over time going back to May 2024.
2. A text box listing active storage and elevation on the date before the script was run. Also includes the storage above the 2019 DCP protection elevationh.
3. Horizontal lines showing teh Minimum Drawdown Elevation and the Catastrophic Elevation with buffer volume.

Data wrangling strategy

1. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation's HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html
2. Import the reservoir elevation-volume curves and critical elevations.
3. Filter the reservoir data for monthly Lake Powell storage.
4. Filter the daily reservoir data for the current Lake Powell and Lake Mead storage/elevation

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), "Minimum Drawdown and Catastrophic Elevations for Lake Powell." Utah State University. Logan, Utah. https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/MinimumDrawdownElevations.

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## [1] "Reservoir Data on: June 5 , 2026"
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## # A tibble: 2 x 3
## # Groups:   ResName [2]
##   ResName     Storage `Pool Elevation`
##   <chr>         <dbl>            <dbl>
## 1 Lake Powell    5.71            3528.
## 2 Lake Mead      7.60            1049.
```

Figure 1. Lake Powell Minimum Drawdown and Catastrophic Elevations.